

Airbnb Multi-City Dataset — Data Audit & Initial Exploration

Listings.csv (279,712 rows) & Reviews.csv (5,373,143 rows) | 10 cities

1. Is the raw data complete, accurate, and ready for analysis?

Strong structural integrity: No duplicate rows, no duplicate listing_id values, and every review is linked to a valid listing.

Price not directly comparable across cities: Prices are stored in each city's local currency without a currency code or exchange rate.

Raw median prices vary widely (e.g., ~65 in Rome, ~1,100 in Bangkok, ~2,405 in Cape Town), making cross-city price comparisons unreliable without currency normalization.

High missingness in host and review metrics: Approximately 40–46% of host_response_rate and host_acceptance_rate, and ~33% of all review score fields are missing.

This likely reflects listings with few or no reviews rather than random data loss.

Limited usefulness of district: Around 87% of district values are blank because the field is populated primarily for New York, making it unsuitable for multi-city analysis.

Invalid pricing and capacity values: 113 listings have price = 0, while 85 listings have accommodates = 0. These are likely data-entry errors and should be cleaned before analysis.

Extreme maximum-night values: 9 listings report maximum_nights ≥ 1,000,000 (up to 2.1 billion), indicating placeholder or erroneous values rather than genuine booking policies.

Conclusion: The dataset is structurally reliable and suitable for exploratory and city-level analysis.

However, before performing cross-city comparisons, predictive modeling, or pricing analysis, it requires currency normalization, outlier treatment, and appropriate handling of missing values.

2. Key Numbers at a Glance

Total listings	279,712
Total reviews	5,373,143
Cities covered	10 (Paris, New York, Sydney, Rome, Rio de Janeiro, Istanbul, Mexico City, Bangkok, Cape Town, Hong Kong)
Unique neighbourhoods	~660 across all cities
Listings with zero reviews	86,156 (31%)
Full duplicate rows	0
Duplicate listing IDs	0
Duplicate review IDs	160
Price = 0 (data error)	113 listings
Review date range	2008-11-16 to 2021-03-01

3. Priority Data-Cleaning Actions Before Deeper Analysis

1. Convert 'price' to a common currency (e.g. USD) using each listing's city as the FX key — this is the single most important fix.
2. Drop or flag the 113 price≤0 rows and 85 accommodates=0 rows as invalid listings.
3. Cap/investigate maximum_nights outliers (9 rows ≥ 1,000,000) — likely meant to represent 'no maximum'.
4. Treat missing review-score fields as 'no reviews yet' rather than imputing — cross-check against Reviews.csv review counts.
5. Drop 'district' from cross-city analysis (single-city coverage) or use only for the New York subset.
6. Investigate host_total_listings_count values in the thousands (max 7,235) — verify against a professional/property-management flag.

2. Data Quality Audit — Missing, Inconsistent & Invalid Values

Listings.csv, 279,712 rows x 33 columns

Missing Values by Column (sorted, worst first)

column	missing_count	missing_pct
district	242700	86.77%
host_response_time	128782	46.04%
host_response_rate	128782	46.04%
host_acceptance_rate	113087	40.43%
review_scores_value	91785	32.81%
review_scores_location	91775	32.81%
review_scores_checkin	91771	32.81%
review_scores_accuracy	91713	32.79%
review_scores_communication	91687	32.78%
review_scores_cleanliness	91665	32.77%
review_scores_rating	91405	32.68%
bedrooms	29435	10.52%
host_location	840	0.30%
host_total_listings_count	165	0.06%
name	175	0.06%
host_is_superhost	165	0.06%
host_since	165	0.06%
host_identity_verified	165	0.06%
host_has_profile_pic	165	0.06%
listing_id	0	0.00%
longitude	0	0.00%
host_id	0	0.00%
latitude	0	0.00%
city	0	0.00%
neighbourhood	0	0.00%
maximum_nights	0	0.00%
minimum_nights	0	0.00%
price	0	0.00%
amenities	0	0.00%
accommodates	0	0.00%
room_type	0	0.00%
property_type	0	0.00%
instant_bookable	0	0.00%

Interpretation

district (86.8% missing): only populated for New York — not a general data-quality defect, a single-city field.

host_response_time / rate (46.0%) & acceptance_rate (40.4%): missing for hosts who have never had a booking request — expected, not an error.

All 7 review_scores_* fields (~32.7-32.8% missing): missing together, consistent with listings that have received no reviews yet.

bedrooms (10.5% missing): likely studio/shared-room listings where bedroom count wasn't specified — needs a business rule (e.g. treat as studio = 0 or 1).

host_location/name/host_since etc. (<0.3% missing): negligible, safe to drop or leave blank.

Review Score Field Scales (inconsistency check)

field	min	max	note
review_scores_rating	20	100	0-100 scale
review_scores_accuracy	2	10	0-10 scale
review_scores_cleanliness	2	10	0-10 scale
review_scores_checkin	2	10	0-10 scale
review_scores_communication	2	10	0-10 scale
review_scores_location	2	10	0-10 scale
review_scores_value	2	10	0-10 scale

Note: review_scores_rating uses a 0-100 scale while the six sub-scores (accuracy, cleanliness, etc.) use a 0-10 scale — a real inconsistency to normalize before combining these fields in any composite score.

2. Duplicate & Referential Integrity Checks

Listings.csv + Reviews.csv cross-check

Integrity Checks

Check	Result
Total listing rows	279712
Duplicate listing_id rows	0
Full duplicate rows (all columns)	0
Total review rows	5373143
Duplicate review_id rows	160
Listings with zero reviews	86156
Review rows referencing unknown listing_id (orphans)	0
Listings with price <= 0	113
Listings with accommodates == 0	85
Rows where minimum_nights > maximum_nights	1
Rows with maximum_nights >= 1,000,000 (data-entry errors)	9
district column populated only for	New York (other cities: 100% blank)

Duplicate/Anomaly Detail

0 fully-duplicated listing rows and 0 duplicate listing_id values — the listings table is clean on the primary key.

160 duplicate review_id values in Reviews.csv (out of 5.37M) — a negligible 0.003%, but worth de-duplicating before counting reviews per listing.

0 orphaned reviews — every review_id references a listing_id present in Listings.csv, confirming referential integrity between the two files.

86,156 listings (30.8%) have never received a review — expected for new/inactive listings, but explains the ~33% missing review-score fields in Sheet 2.

1 listing has minimum_nights > maximum_nights (a logical impossibility) and 9 listings show maximum_nights of 1M+ (up to 2.1 billion) — both are placeholder/data-entry errors, not real host policies.

Hosts with Unusually High Total Listings Count (possible property-management companies)

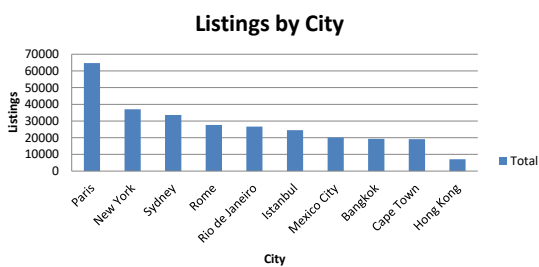
host_id	city	host_total_listings_count
175128252	Sydney	7235

host_total_listings_count median across dataset = 1; this host manages 7,235 — almost certainly a professional property-management operation rather than an individual, and should be segmented separately in host-level analysis.

3. Listing Distribution — Cities, Neighbourhoods, Property & Room Types

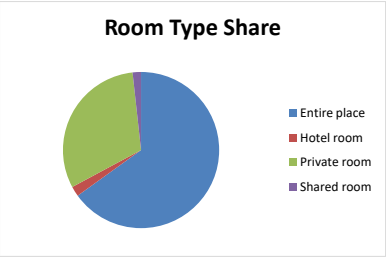
Listings by City

city	Entire place	Hotel room	Private room	Shared room	Total
Paris	55378	1396	7492	424	64690
New York	19397	299	16630	686	37012
Sydney	20358	233	12445	594	33630
Rome	17248	1276	8923	200	27647
Rio de Janeiro	19285	99	6619	612	26615
Istanbul	12429	802	10605	683	24519
Mexico City	10548	246	8890	381	20065
Bangkok	10623	1020	6993	725	19361
Cape Town	14150	306	4503	127	19086
Hong Kong	2589	180	3888	430	7087



Room Type Split (All Cities)

room_type	count	pct
Entire place	182005	65.1%
Hotel room	5857	2.1%
Private room	86988	31.1%
Shared room	4862	1.7%



Top Property Types (All Cities)

property_type	count	pct
Entire apartment	138989	49.69%
Private room in apartment	47322	16.92%
Private room in house	13292	4.75%
Entire house	13273	4.75%
Entire condominium	11250	4.02%
Room in boutique hotel	5771	2.06%
Entire loft	4587	1.64%
Private room in condominium	4462	1.60%
Private room in bed and breakfast	4238	1.52%
Entire serviced apartment	3973	1.42%
Room in hotel	3205	1.15%
Private room in townhouse	2959	1.06%

Top 10 Neighbourhoods per City (by listing count)

city	neighbourhood	count
Bangkok	Vadhana	2455
Bangkok	Khlong Toei	2376
Bangkok	Huai Khwang	1500
Bangkok	Ratchathewi	1371
Bangkok	Bang Rak	1079
Bangkok	Sathon	880
Bangkok	Phra Nakhon	802
Bangkok	Phra Khanong	696
Bangkok	Chatu Chak	632
Bangkok	Din Daeng	576
Cape Town	Ward 115	3293
Cape Town	Ward 54	2562
Cape Town	Ward 77	1227
Cape Town	Ward 23	1056
Cape Town	Ward 74	981
Cape Town	Ward 64	944
Cape Town	Ward 61	824
Cape Town	Ward 62	612
Cape Town	Ward 59	549
Cape Town	Ward 107	528
Hong Kong	Yau Tsim Mong	3038
Hong Kong	Wan Chai	1438
Hong Kong	Central & Western	930
Hong Kong	Kowloon City	332
Hong Kong	Islands	289
Hong Kong	Eastern	265
Hong Kong	North	193
Hong Kong	Sham Shui Po	154
Hong Kong	Yuen Long	132
Hong Kong	Sai Kung	84

Istanbul	Beyoglu	6674
Istanbul	Sisli	3522
Istanbul	Fatih	3326
Istanbul	Kadikoy	2260
Istanbul	Besiktas	1632
Istanbul	Uskudar	727
Istanbul	Esenyurt	720
Istanbul	Kagithane	662
Istanbul	Atasehir	397
Istanbul	Maltepe	386
Mexico City	Cuauhtemoc	7626
Mexico City	Miguel Hidalgo	3288
Mexico City	Benito Juarez	3133
Mexico City	Coyoacan	1675
Mexico City	Alvaro Obregon	977
Mexico City	Tlalpan	841
Mexico City	Cuajimalpa de Morelos	516
Mexico City	Venustiano Carranza	440
Mexico City	Gustavo A. Madero	386
Mexico City	Iztapalapa	300
New York	Williamsburg	2733
New York	Bedford-Stuyvesant	2711
New York	Harlem	1972
New York	Bushwick	1678
New York	Hell's Kitchen	1506
New York	Upper West Side	1454
New York	Midtown	1451
New York	Upper East Side	1342
New York	East Village	1314
New York	Crown Heights	1127
Paris	Buttes-Montmartre	7237
Paris	Popincourt	6206
Paris	Vaugirard	4826
Paris	Enclos-St-Laurent	4628
Paris	Batignolles-Monceau	4330
Paris	Menilmontant	3758
Paris	Buttes-Chaumont	3728
Paris	Passy	3215
Paris	Opera	3157
Paris	Temple	2943
Rio de Janeiro	Copacabana	7712
Rio de Janeiro	Barra da Tijuca	2724
Rio de Janeiro	Ipanema	2572
Rio de Janeiro	Jacarepagua	1335
Rio de Janeiro	Botafogo	1222
Rio de Janeiro	Recreio dos Bandeirantes	1179
Rio de Janeiro	Leblon	1165
Rio de Janeiro	Santa Teresa	882
Rio de Janeiro	Centro	773
Rio de Janeiro	Flamengo	612
Rome	I Centro Storico	14874
Rome	VII San Giovanni/Colonna	2079
Rome	II Parioli/Nomentano	1991
Rome	XIII Aurelia	1697
Rome	XII Monte Verde	1323
Rome	V Prenestino/Centurio	1087
Rome	VIII Appia Antica	804
Rome	X Ostia/Aclia	631
Rome	XIV Monte Mario	573
Rome	IV Tiburtina	532
Sydney	Sydney	8074
Sydney	Waverley	4737
Sydney	Randwick	3045
Sydney	Warringah	1705
Sydney	Manly	1631
Sydney	Woollahra	1382
Sydney	North Sydney	1188
Sydney	Pittwater	1165
Sydney	Marrickville	1111
Sydney	Rockdale	855

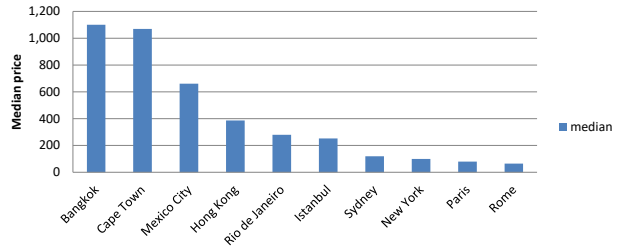
4. Initial Pricing Patterns Across Markets

NOTE: price is in each city's LOCAL CURRENCY (no FX conversion applied)

Price Summary by City (local currency, raw)

city	count	mean	median	std	min	max
Bangkok	19361	2,078	1,100	6,380	0	3,00,177
Cape Town	19086	2,405	1,069	6,143	122	1,80,000
Mexico City	20065	1,149	661	5,372	0	4,99,000
Hong Kong	7087	746	386	1,938	1	83,665
Rio de Janeiro	26615	743	280	5,369	0	6,25,216
Istanbul	24519	533	252	3,318	0	1,79,532
Sydney	33630	222	120	473	5	28,613
New York	37012	143	99	276	0	10,000
Paris	64690	113	80	214	0	12,000
Rome	27647	105	65	341	0	10,571

Median Price by City (local currency)



Median Price by City x Room Type (local currency)

city	Entire place	Hotel room	Private room	Shared room
Bangkok	1192	1271	1000	450
Cape Town	1288	1350	622	282
Hong Kong	690	450	300	186
Istanbul	350	324	163	100
Mexico City	950	1204	400	300
New York	140	150	60	45
Paris	82	158	53	34
Rio de Janeiro	350	183	160	100
Rome	77	60	49	20
Sydney	176	149	68	39

Reading These Numbers

Bangkok (THB) and Cape Town (ZAR) record the highest raw median prices (~1,100 and ~1,069, respectively).

However, this is primarily due to their local currency denominations rather than higher accommodation costs.

Rome (EUR) and Paris (EUR) display the lowest raw median prices (~65–80).

This reflects the higher value of the Euro relative to other currencies and should not be interpreted as these cities being the cheapest Airbnb markets.

Cross-city price comparisons are not valid using the raw price field because each city is reported in its own local currency without exchange rate normalization.

Within-city pricing comparisons remain reliable. Across all cities, Entire Place listings consistently command higher prices than Private Room and Shared Room listings.

Since these comparisons are made within the same currency, the pricing hierarchy is unaffected by exchange rate differences.

5. Outlier & Anomaly Detection

Price, ratings, and host-info anomalies (IQR method: outlier = below Q1-1.5xIQR or above Q3+1.5xIQR)

Price Outliers by City (IQR method, local currency)

city	n_listings	iqr_lower_bound	iqr_upper_bound	n_price_outliers	pct_outliers
Rio de Janeiro	26615	-433	1,140	2,770	10.41
Cape Town	19086	-1,675	4,525	1,971	10.33
Sydney	33630	-144	437	3,328	9.9
Hong Kong	7087	-430	1,378	620	8.75
Bangkok	19361	-1,100	3,700	1,680	8.68
Rome	27647	-36	180	2,232	8.07
Paris	64690	-33	212	5,164	7.98
Istanbul	24519	-299	898	1,917	7.82
Mexico City	20065	-709	2,224	1,497	7.46
New York	37012	-77	288	2,727	7.37

7-10% of listings in every city fall outside the IQR fence — driven by a long right tail of luxury villas, hotel rooms, and likely mispriced/placeholder listings, not by any single data error.

Top 20 Highest-Priced Listings (raw, local currency — sanity check candidates)

listing_id	city	property_type	room_type	price	accommodates
13879989	Rio de Janeiro	Shared room in house	Shared room	6,25,216	5
47885441	Mexico City	Room in boutique hotel	Private room	4,99,000	4
23895520	Mexico City	Private room in house	Private room	3,50,000	4
3410763	Bangkok	Private room in apartm	Private room	3,00,177	2
13033135	Bangkok	Private room in house	Private room	3,00,000	4
12056763	Bangkok	Entire house	Entire place	3,00,000	4
29970931	Bangkok	Entire apartment	Entire place	2,31,047	2
38733276	Mexico City	Private room in apartm	Private room	2,06,499	1
41578696	Bangkok	Entire villa	Entire place	1,80,124	10
24231503	Cape Town	Entire villa	Entire place	1,80,000	10
41615363	Mexico City	Entire house	Entire place	1,80,000	4
36421447	Cape Town	Entire villa	Entire place	1,80,000	10
14658972	Istanbul	Private room in bed and	Private room	1,79,532	2
16111771	Istanbul	Room in boutique hotel	Private room	1,79,532	2
16111705	Istanbul	Room in boutique hotel	Private room	1,79,532	2
16401723	Istanbul	Room in boutique hotel	Private room	1,79,153	2
23920872	Mexico City	Entire apartment	Entire place	1,63,502	8
16045517	Bangkok	Entire apartment	Entire place	1,60,000	4
29391240	Cape Town	Entire villa	Entire place	1,57,500	10
39296889	Cape Town	Entire villa	Entire place	1,55,000	10

Several of these (e.g. 625,216 in Rio; 499,000 in Mexico City) are 100-1000x their city's median and warrant manual verification — likely fat-finger entries (extra zero) rather than genuine nightly rates.

Zero/Invalid Price Listings by City

city	count_zero_price
Bangkok	1
Istanbul	2
Mexico City	7
New York	28
Paris	62
Rio de Janeiro	7
Rome	6

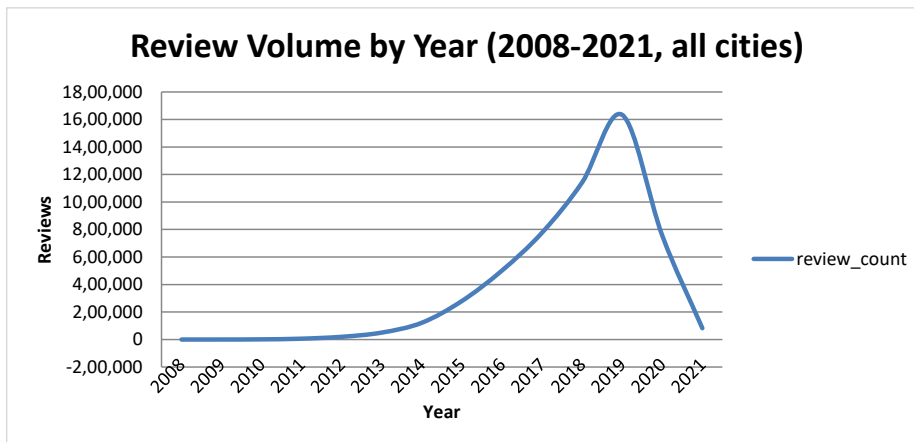
Other Anomalies Flagged

Ratings scale mismatch: review_scores_rating runs 0-100 while all six review_scores_* sub-metrics run 0-10 — must be rescaled before combining. review_scores_rating minimum observed is 20 (not 0) and sub-scores bottom out at 2 (not 0) — suggests Airbnb's own scale floor, not missing-data placeholders. host_total_listings_count ranges from 0 to 7,235 (median = 1) — the extreme values belong to a small number of professional hosts/property managers. 1 listing has minimum_nights > maximum_nights, and 9 listings show maximum_nights >= 1,000,000 (max 2,147,483,647 — the classic 32-bit integer overflow placeholder value) — treat these as 'no maximum specified' rather than a real nightly cap. host_response_rate and host_acceptance_rate are already stored as clean 0-1 fractions with no out-of-range values — no rescaling needed there.

6. High-Level Trends Before Deeper Analysis

Review Volume by Year (proxy for market growth / demand)

year	review_count
2008	2
2009	115
2010	1,236
2011	6,253
2012	19,922
2013	50,522
2014	1,22,132
2015	2,80,332
2016	5,01,754
2017	7,77,200
2018	11,42,496
2019	16,33,546
2020	7,55,324
2021	82,309



Key Takeaways

1. Market composition: 'Entire place' listings dominate (65% of all listings), with Paris (64,690), New York (37,012) and Sydney (33,630) the three largest markets by listing count — together ~48% of the dataset.
2. Host structure is highly individual: median host_total_listings_count is 1, but a thin layer of professional multi-property hosts (up to 7,235 listings) exists and should be segmented out of 'typical host' analysis.
3. Review activity grew steadily from 2008 through ~2019 before a sharp dip in the final data-collection year (2020/2021 cut short mid-year) — consistent with the scrape simply ending in early 2021 rather than a real demand collapse; confirm the exact scrape date before reading anything into the 2020-21 numbers.
4. Nearly a third of listings (31%) have zero reviews — a substantial 'cold start' segment that will bias any review-score-based analysis unless handled separately (e.g. flag as 'new/unreviewed' rather than imputing scores).
5. Pricing cannot be compared across cities in its current form (see Sheet 5) — this is the top blocker for any multi-city pricing or elasticity work.
6. Superhost rate is 18% (50,253 of 279,547 with known status) and instant-bookable listings are 41% (115,607) — both reasonable, plausible splits with no red flags.
7. Neighbourhood granularity varies enormously by city (New York: 220 neighbourhoods; Mexico City: 16) — comparisons at the neighbourhood level should therefore be done within a city, not across cities.